

Q29 Ans: C

Q30 Ans B

Let the initial count rate due to source only be N .

After 2 half-lives, the count rate will drop by $\frac{3}{4}N$.

Hence,

$$\frac{3}{4}N = 324 - 108$$

$$N = 288$$

$$\text{Background radiation} = 324 - 288 = 36$$